

Hybrid III 6 y/o Lab

BME254L - Spring 2026 - Palmeri

Dr. Mark Palmeri, M.D., Ph.D.

2026-01-07

Table of contents

Tasks	1
What to submit	2
Bottom Plate	4
Top Plate	6
Nodding Plate	8
Molded Neck	10
Lower Neck Bracket	13
Load Cell	14
Assembly	15

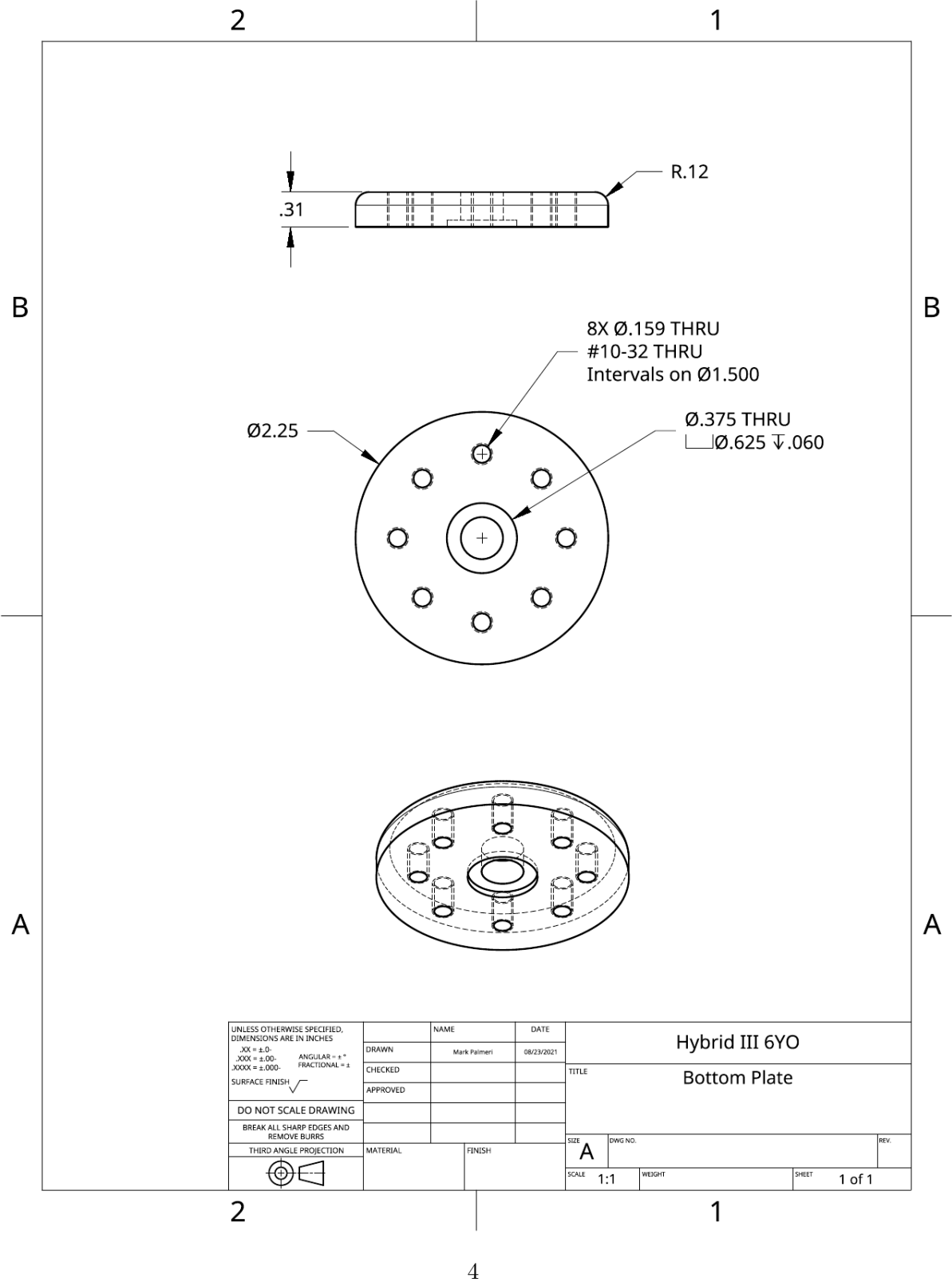
Tasks

- Create all of the parts for the Hybrid III 6yo test setup based on the attached drawings.
- You should create all of these parts and their associated drawings and assemblies as part of the same “Document” and share that document with Dr. Palmeri and the TA(s).
- Create the mechanical drawings associated with each of those parts (i.e., recreate the drawings that you are using to generate the parts) and the assembly.
- Assemble your parts as guided by the assembly diagram and as shown in the uploaded animation.
- There are online tutorials on Assemblies if you need some refreshers:
 - <https://learn.onshape.com/courses/introduction-to-assembly-design>
 - <https://learn.onshape.com/courses/fundamentals-onshape-assemblies>

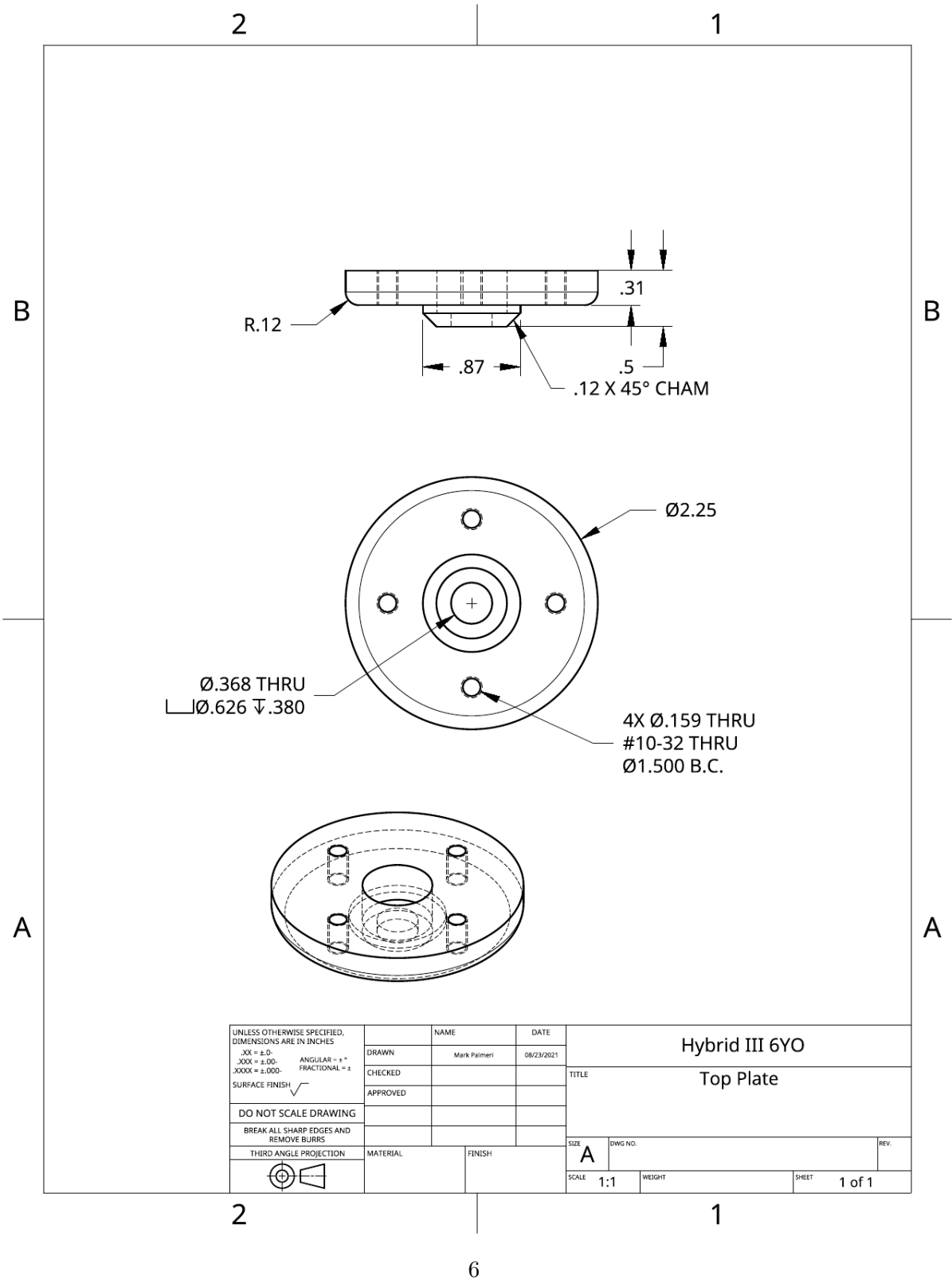
What to submit

Upload mechanical drawings and a screenshot of the part for all of your parts to the Hybrid III 6 y/o Gradescope assignment.

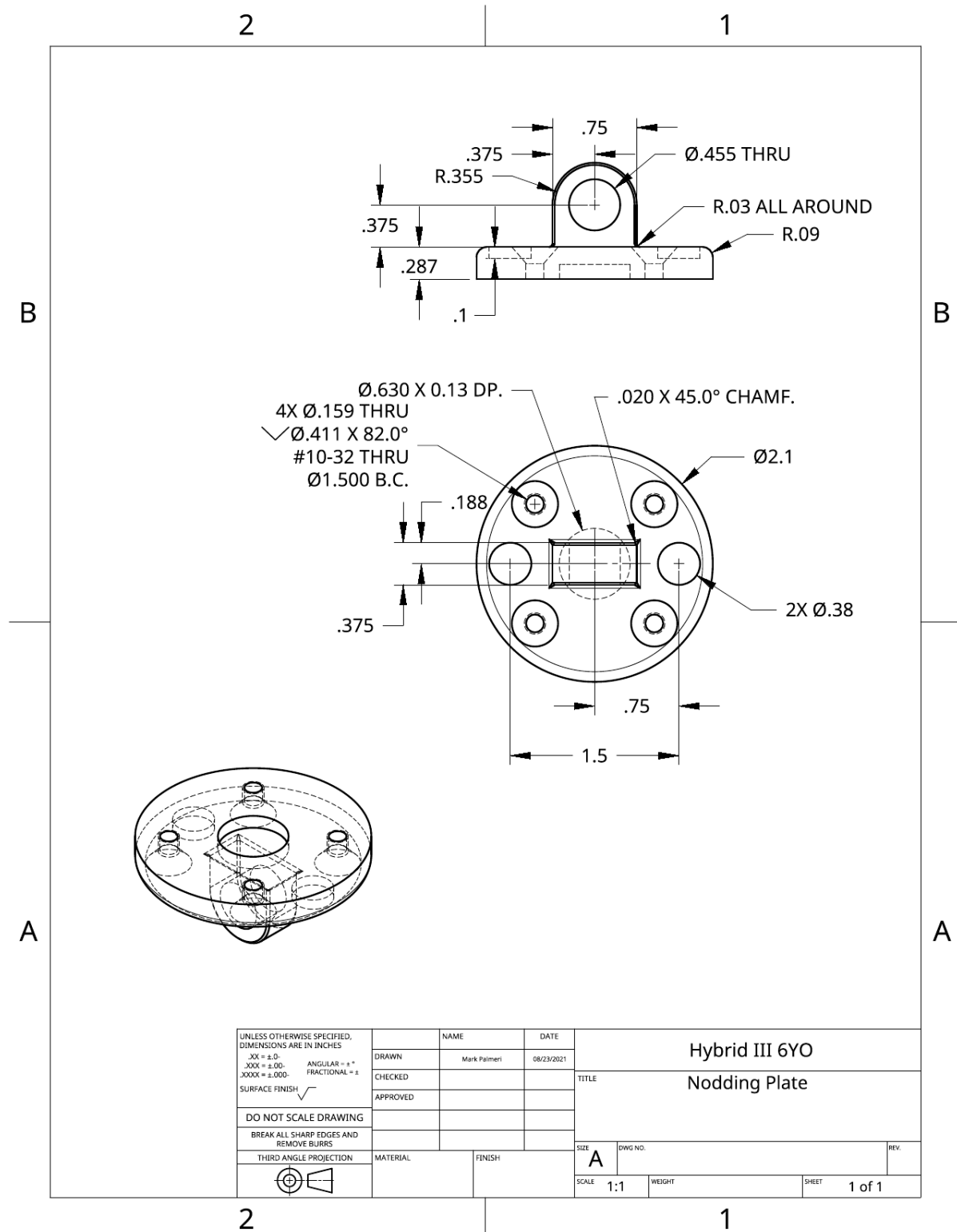
Bottom Plate



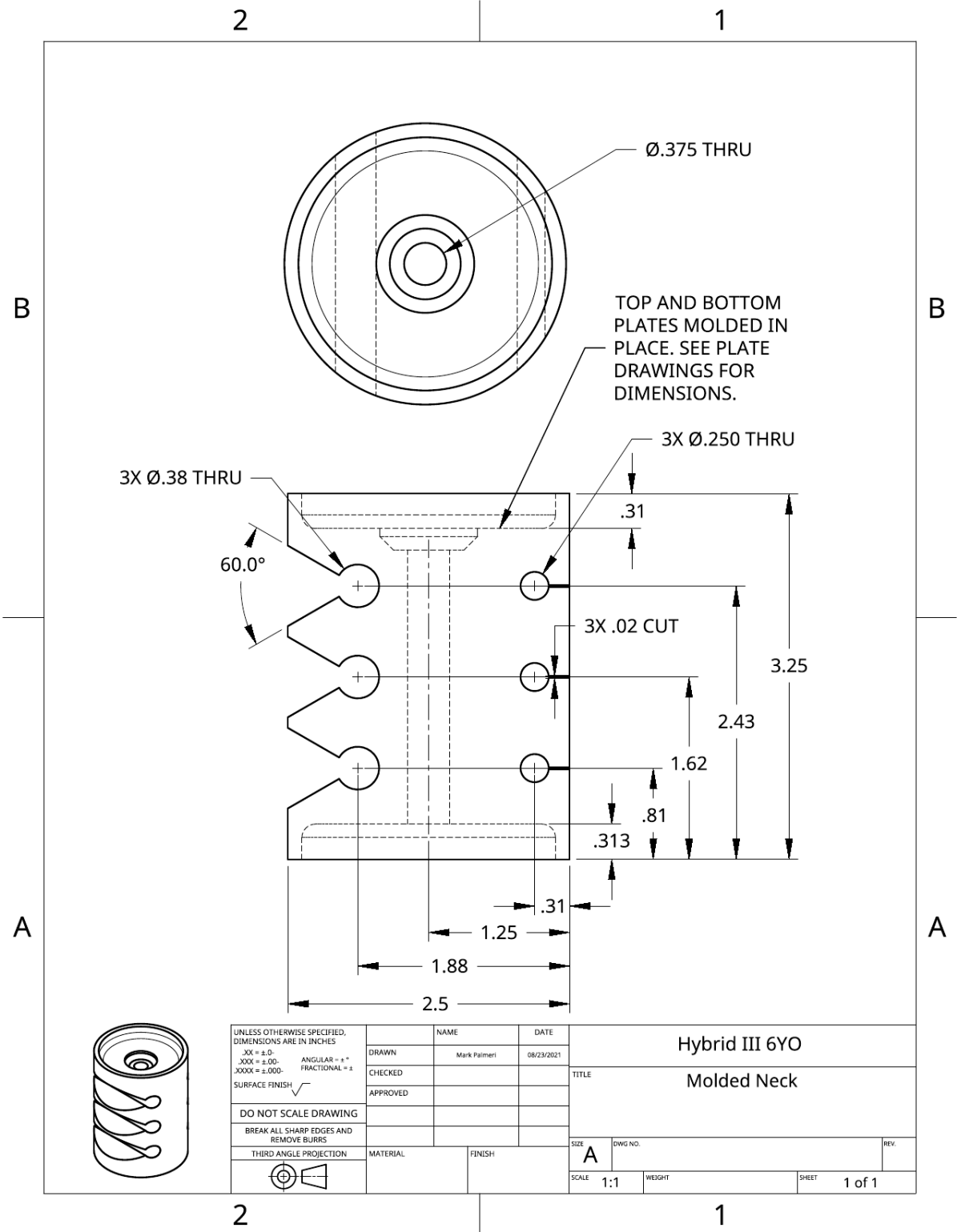
Top Plate



Nodding Plate



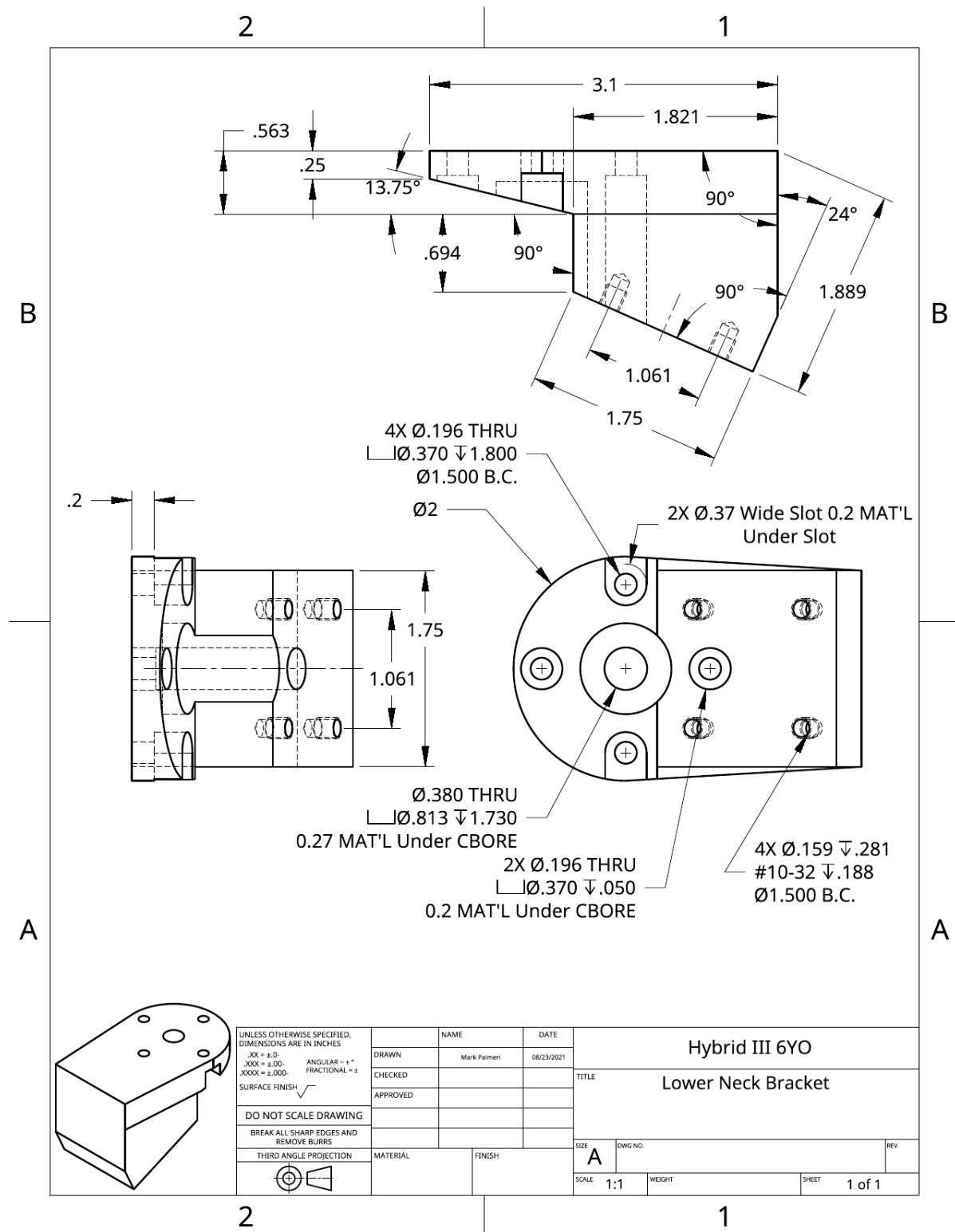
Molded Neck



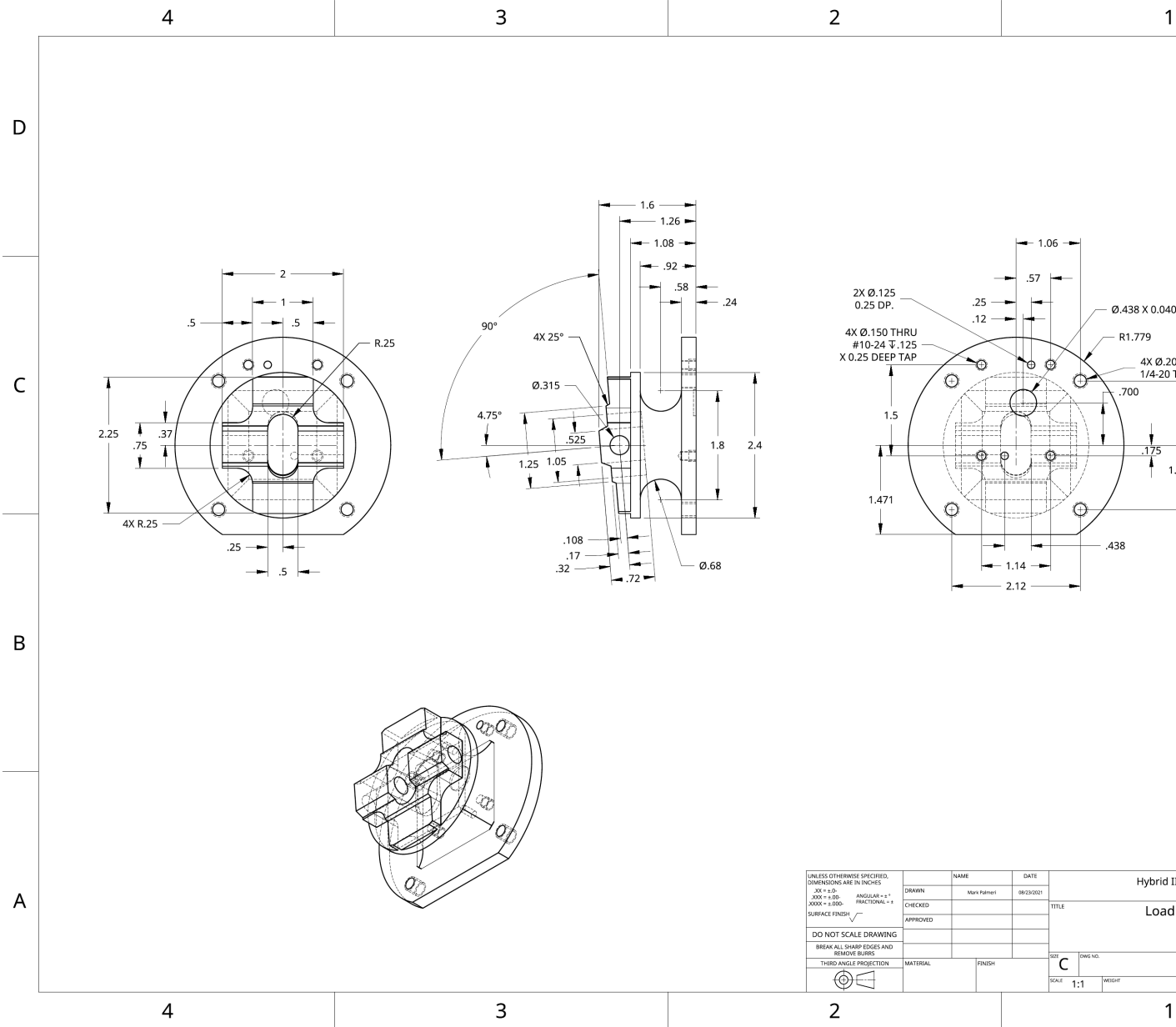
 Tip

Note that the this part has features that are dependent on the top and bottom plates. To reduce the overhead of manually entering these dimensions from the other parts, you can use the **Derived** tool in Onshape to bring in the relevant faces from the other parts. <https://cad.onshape.com/help/Content/derived.htm#StepsDesktop>

Lower Neck Bracket



Load Cell



SECTION A - A
SCALE 1:2

SCALE 1:2

SCALE 1:2

Item	Quantity	Part number	Material
1	1	MOLDED NECK	Butyl
2	1	TOP PLATE	Steel
3	1	NODDING PLATE	Steel
4	1	BOTTOM PLATE	Steel
5	1	LOWER NECK BRACKET	Aluminum

UNLESS OTHERWISE SPECIFIED,
DIMENSIONS ARE IN INCHES:
 .XX = ±.01
 .XXX = ±.005
 .XXXX = ±.0005
 ANGULAR = ±.5°
 FRACTIONAL = ±.005

SURFACE FINISH $\sqrt{\text{ }}$

DO NOT SCALE DRAWING

BREAK ALL SHARP EDGES AND
REMOVE BURRS

THIRD ANGLE PROJECTION

NAME
DATE
DRAWN
CHECKED
APPROVED

MATERIAL
FINISH

SIZE
SCALE
1:1

DWG NO.
B

WEIGHT

Hybrid III 6YC
Assembly